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Give **two** structural features of an aorta wall and explain how they are related to the function of an aorta.

[2 marks]

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2 \_\_\_\_\_

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**Question 8 continues on the next page**

0 8 . 2

Small tears may occur in the layers of tissue in an aorta wall. These tears weaken the wall without bursting the aorta.

Scientists:

- measured the aorta diameter ( $d$ ) in a large population of people over 60 years of age
- calculated the risk of an aorta wall developing a tear.

**Table 4** shows their results.

**Table 4**

| Aorta diameter $d$ / cm | Number of people in the population | Risk of an aorta wall developing a tear |
|-------------------------|------------------------------------|-----------------------------------------|
| $d \leq 3.5$            | 2765                               | 0.06                                    |
| $3.5 < d \leq 4.0$      | 630                                | 0.33                                    |
| $4.0 < d \leq 4.5$      | 98                                 | 4.64                                    |
| $d > 4.5$               | 7                                  | 380.00                                  |

Blood may push through the tears in the aorta wall. This produces a balloon-like swelling called an aneurysm and increases the aorta diameter. Aneurysms can cause the aorta to burst.

Using all the information, what can you conclude about aorta diameter and the risk of developing an aneurysm?

**[3 marks]**

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0 8 . 3

A scientist investigated changes in a diseased heart and changes in a healthy heart during cardiac cycles.

For each heart, the scientist obtained a value for:

- the mean blood volume in a full ventricle just before the ventricle contracts (**BVB**)
- the mean ejection fraction (**EF**).

The **EF** is the proportion of blood pumped out of a full ventricle in one heartbeat.

The **EF** is calculated using this formula:

$$EF = \frac{\text{Blood volume pumped out of a full ventricle in one heartbeat (stroke volume)}}{BVB}$$

**Table 5** shows the scientist's results.

**Table 5**

|                                   | Diseased heart | Healthy heart |
|-----------------------------------|----------------|---------------|
| Mean <b>BVB</b> / cm <sup>3</sup> | 100            | 120           |
| Mean <b>EF</b>                    | 0.45           | 0.58          |

Using **Table 5**, a student calculated that the percentage change in the stroke volume of the diseased heart compared with the stroke volume of a healthy heart is –30%.

The student's answer is wrong because the final step of the calculation was performed incorrectly.

Using the equation and **Table 5**, calculate the correct percentage change in the stroke volume of the diseased heart compared with the stroke volume of the healthy heart.

Identify the mathematical error in the final step of the student's calculation.

**[3 marks]**

Correct answer \_\_\_\_\_ %

Mathematical error \_\_\_\_\_

